

Final Project Description

The overall goal of this final project is to use data science to investigate an urban challenge and design a research study around it. In groups of 1-2, you will design a research question about your challenge, formulate hypotheses, and analyze data to advance our understanding of the challenge.

The study must include the following elements:

- Two or more datasets
- One or more methods from class
- Produce three or more “outputs” (figures, tables) (GR students: four or more outputs)

And:

- A testable research question around your urban challenge
- Transformation of data (i.e., more than simply descriptive statistics)
- Interpretation of your results: what is the answer to your question given the evidence?
- Implications of your findings. How might these results inform policymakers, community organizers, or researchers in urban studies?

There will be four main deliverables:

1. Proposal

Halfway through the term, students will turn in a proposal for their final project. This will include a research question, justification for the study, proposed data and methodology, and hypothesis about findings. It is expected that this may change in some ways before the end of the term, but this will help guide students in building the final project.

2. Presentation

Prepare a ~10 minute presentation on your study to be shared with the class during the last two class days. This is your opportunity to tell a story about your challenge, your data, and your findings. Be creative! The presentation must include the research question, succinct explanation of data and methods, the study outputs, and discussion on implications of the study.

3. Report and reflection

After the presentation, you will put together a written report with more details about your study. The same criteria for the presentation elements should be in the report. The first part of the report will be **10 pages maximum (GR: 11 pages maximum)** and can be designed as an article or a report. The report should have the following sections:

Motivation (UG students only)

4 paragraphs:

- explain the real-world context and motivation of your study
- review relevant previous work to establish what is currently known and identify what important open question remains (gap in the literature or unmet need in practice)
- problem and its significance, open RQ, what argument developed here, why important
- roadmap of the report

Introduction (GR students only)

4 paragraphs:

- background, context, motivation
- problem and its significance, open RQ, what argument developed here, why important
- how did you answer your question and what did you find? Data, methods, findings
- roadmap of the report

Background (GR students only)

- explain the real-world context of your study
- review relevant previous work to establish what is currently known
- identify what important open question remains (gap in the literature or unmet need in practice)

Methods

state your research question and hypotheses

data subsection: describe your data collection, variables, and any relevant processing

analysis subsection: describe your analysis with enough detail that an expert could replicate it

Findings

what did your analysis reveal: organize your results around your argument but present them objectively with limited interpretation

include supporting tables/figures

do not mix in any methods

Discussion

answer your question, advance your argument, and demonstrate significance: return to research question and interpret (don't repeat) your findings as evidence for/against it

tell a story: link your findings together to persuasively advance your argument

significance: discuss specific implications for research and practice ("so what")

acknowledge study's limitations and alternative interpretations of your evidence

Conclusion

succinctly wrap up

summarize your topic, question, and argument

summarize what you did, what you found, what it means, and why it matters

After the presentations, you will also add a ½ **page reflection** on the final project itself and the projects of your classmates. Respond to the following prompt:

Reflect on the process of gathering data, identifying the problem, the implications, etc. What was challenging? Easy? Were you surprised by the process? Reflect on the other presentations from your classmates. How were they different? The same? What themes emerged across the class? What did you learn after all of the presentations? How do the other submissions relate to the course readings and material?

Upload the report and reflection to Canvas as **one PDF**.

4. Code for Project

Provide your code, clean and commented, in HTML format.